Sports Multiple 36 Kids - Printable PDF

Manufactured by QRU Factory(TM). QRU simplifies the path to understanding the truth.

FRONT MATTER

Product Governance Package(TM)

- This material is educational and reflects governed knowledge at time of manufacture.

Manufactured by the QRU Factory(TM) from a Verified Knowledge Record(TM). AI assisted drafting and advisory scoring; all governed standards were checked by the Factory. AI advisory never approves, certifies, or publishes - governance authority is human/constitutional.

Accessibility: Plain-language explanation, defined vocabulary, and alt-text for visuals where applicable.

Table of Contents

QRU Sports Intelligence Lab(TM)

CHAPTER 1

36 Sports Multiplication Challenges for Kids

1.1 Train Your Brain Like an Athlete Trains Their Body(TM)

Audience: General Public
Learning Level: Introductory
Purpose: Help children build multiplication understanding, confidence, reasoning, and problem-solving through sports-based practice.

CHAPTER 2

Printable PDF Overview

Multiplication can feel easier when it connects to something children already understand: sports.

In sports, athletes practice skills over and over. They miss shots, drop passes, lose points, try again, listen to coaches, adjust their strategy, and improve.

Learning math works the same way.

A child is not "bad at math" because they miss one problem. A missed problem is a training rep for the brain.

1. The Big Idea

CHAPTER 3

Children Do Not Need More Information

CHAPTER 4

They Need Stronger Understanding Systems

QRU Sports Intelligence Lab(TM) turns multiplication practice into athlete-style brain training.

Instead of learning multiplication only by memorizing facts, children learn to see multiplication as:

- Groups
- Patterns
- Repeated practice
- Strategy
- Teamwork
- Real-world problem solving

The learner becomes a Brain Athlete.

---

2. What Is Multiplication?

CHAPTER 5

Simple Definition

Multiplication means many equal groups combined.

Example:

If a basketball player makes 4 shots in each round and plays 3 rounds, that means:

$4 + 4 + 4 = 12$

Or:

$4 \times 3 = 12$

So, multiplication is a faster way to count equal groups.

CHAPTER 6

Vocabulary Decoder(TM)

6.1 Multiplication

Multi = many
Groups = sets of things

Multiplication means many groups combined.

6.2 Intelligence

Intelligence is the ability to:

- Learn
- Understand
- Adapt
- Solve problems

- Keep improving

3. Everyday Analogy

Learning math is like practicing a sport.

One missed shot does not mean you are bad at basketball.

One wrong answer does not mean you are bad at math.

Both are training reps.

Each try helps your brain get stronger.

4. The Brain Athlete Mindset

A Brain Athlete says:

- "I can practice."
- "Mistakes help me improve."
- "I can try a different strategy."
- "Understanding creates confidence."
- "I do not have to be perfect to get better."

5. What Multiplication Is NOT

Multiplication is not:

- A race to prove who is smartest
- Memorizing without understanding

- A reason to feel embarrassed
- A test of your worth
- Something only "math people" can learn

Multiplication is a skill.

Skills grow through practice.

6. Coach Notes for Parents, Teachers, and Caregivers

Your role is to be the child's coach.

A good coach does not shame an athlete for missing.

A good coach helps the athlete notice, adjust, and try again.

When a child struggles, try saying:

- "Let's look at the groups."
- "What strategy could we try?"
- "What do you already know?"
- "Show me how you thought about it."
- "That mistake gave us information."

The goal is not perfection.

The goal is helping the child discover:

CHAPTER 7

"I can learn difficult things."

7. How to Use This Workbook

For each challenge:

1. Read the sports story.
2. Find the number of groups.
3. Find how many are in each group.
4. Write the multiplication sentence.
5. Solve.
6. Explain your thinking.

Example:

A soccer player scores 2 goals in each of 4 games. How many goals total?

Groups: 4 games
Number in each group: 2 goals
Multiplication sentence: $4 \times 2 = 8$
Answer: 8 goals

8. Warm-Up Practice

CHAPTER 8

Draw the Groups

8.1 Problem A

A baseball team practices 3 drills.
Each drill has 5 throws.

Draw 3 groups of 5.

Multiplication sentence:

3 × 5 =

Answer:

8.2 Problem B

A swimmer swims 4 laps each day for 2 days.

Draw 2 groups of 4.

Multiplication sentence:

2 × 4 =

Answer:

---

9. The 36 Sports Multiplication Challenges

CHAPTER 9

Football Challenges

9.1 1. Quarterback Passes

A quarterback completes 5 passes in each quarter.

There are 4 quarters.

5 × 4 =

Answer:

9.2 2. Touchdown Practice

A player practices 3 touchdown runs each day for 6 days.

3 × 6 =

Answer:

9.3 3. Team Huddles

A football team has 7 players in each small huddle.
There are 3 huddles.

7 × 3 =

Answer:

9.4 4. Field Goal Kicks

A kicker makes 4 field goals during each practice.
There are 5 practices.

4 × 5 =

Answer:

CHAPTER 10

Basketball Challenges

10.1 5. Free Throws

A basketball player makes 8 free throws each practice.
There are 5 practices.

$8 \times 5 =$

Answer:

10.2 6. Three-Point Shots

A player makes 3 three-point shots in each game.
They play 4 games.

$3 \times 4 =$

Answer:

10.3 7. Team Layups

Each player makes 6 layups.
There are 5 players.

$6 \times 5 =$

Answer:

10.4 8. Basketball Drills

A team does 4 drills.
Each drill lasts 7 minutes.

$4 \times 7 =$

Answer:

CHAPTER 11

Soccer Challenges

11.1 9. Soccer Goals

A player scores 2 goals in each game.
They play 6 games.

$2 \times 6 =$

Answer:

11.2 10. Passing Practice

A soccer player makes 9 passes in each round.
There are 3 rounds.

$9 \times 3 =$

Answer:

11.3 11. Team Cones

A coach places 5 cones in each line.
There are 6 lines.

$5 \times 6 =$

Answer:

11.4 12. Penalty Kicks

A player takes 4 penalty kicks each day for 4 days.

$4 \times 4 =$

Answer:

CHAPTER 12

Baseball and Softball Challenges

12.1 13. Batting Practice

A player hits 6 balls in each round.
There are 4 rounds.

$6 \times 4 =$

Answer:

12.2 14. Baseball Gloves

There are 9 players on a team.
Each player has 1 glove.

$9 \times 1 =$

Answer:

12.3 15. Softball Throws

A player makes 7 throws each inning for 3 innings.

$7 \times 3 =$

Answer:

12.4 16. Home Run Cards

A collector gets 5 baseball cards each week for 8 weeks.

$5 \times 8 =$

Answer:

CHAPTER 13

Track and Running Challenges

13.1 17. Running Laps

A runner completes 4 laps each day for 5 days.

$4 \times 5 =$

Answer:

13.2 18. Sprint Practice

A sprinter runs 6 sprints per practice.
There are 3 practices.

$6 \times 3 =$

Answer:

13.3 19. Relay Teams

Each relay team has 4 runners.
There are 6 teams.

$4 \times 6 =$

Answer:

13.4 20. Jumping Practice

An athlete does 8 jumps in each round.
There are 2 rounds.

$8 \times 2 =$

Answer:

Swimming Challenges

14.1 21. Pool Laps

A swimmer swims 5 laps each morning for 6 mornings.

$5 \times 6 =$

Answer:

14.2 22. Swim Kicks

A swimmer practices 10 kicks in each set.
There are 4 sets.

$10 \times 4 =$

Answer:

14.3 23. Relay Swimmers

Each relay team has 4 swimmers.
There are 3 relay teams.

$4 \times 3 =$

Answer:

14.4 24. Practice Days

A swimmer trains 2 hours each day for 7 days.

$2 \times 7 =$

Answer:

CHAPTER 15

Tennis Challenges

15.1 25. Tennis Serves

A tennis player practices 6 serves each round.
There are 5 rounds.

$6 \times 5 =$

Answer:

15.2 26. Tennis Balls

A coach puts 3 tennis balls in each basket.
There are 8 baskets.

$3 \times 8 =$

Answer:

15.3 27. Practice Matches

A player wins 2 games in each practice match.
They play 6 practice matches.

$2 \times 6 =$

Answer:

15.4 28. Racquet Strings

A player checks 4 racquets.
Each racquet has 7 loose strings to fix.

$4 \times 7 =$

Answer:

CHAPTER 16

Mixed Sports Challenges

16.1 29. Hockey Shots

A hockey player takes 9 shots each period.
There are 3 periods.

$9 \times 3 =$

Answer:

16.2 30. Volleyball Serves

A volleyball player practices 5 serves each round.
There are 7 rounds.

$5 \times 7 =$

Answer:

16.3 31. Golf Swings

A golfer takes 8 practice swings at each hole.
They practice at 3 holes.

$8 \times 3 =$

Answer:

16.4 32. Gymnastics Routines

A gymnast practices 4 routines each day for 6 days.

$4 \times 6 =$

Answer:

16.5 33. Skating Spins

A skater does 3 spins in each routine.
They practice 5 routines.

$3 \times 5 =$

Answer:

16.6 34. Martial Arts Kicks

A martial artist practices 10 kicks in each set.
There are 6 sets.

$10 \times 6 =$

Answer:

16.7 35. Bowling Pins

A bowler knocks down 7 pins in each frame.
They play 4 frames.

$7 \times 4 =$

Answer:

16.8 36. Team Water Bottles

A coach gives 2 water bottles to each player.
There are 9 players.

$2 \times 9 =$

Answer:

10. Strategy Zone

CHAPTER 17

Three Ways to Solve Multiplication

17.1 1. Repeated Addition

Example:

4×3

Think:

$4 + 4 + 4 = 12$

17.2 2. Equal Groups

Example:

There are 3 groups with 4 in each group.

That means:

$3 \times 4 = 12$

17.3 3. Skip Counting

Example:

To solve 5×6, count by 5 six times:

5, 10, 15, 20, 25, 30

So:

$5 \times 6 = 30$

---

11. Reflection: Game Film Review for the Brain

Athletes watch game film to improve.

Brain Athletes reflect on their thinking.

Answer these questions:

1. Which problem felt easiest?

2. Which problem felt hardest?

3. What strategy helped you most?

4. What mistake helped you learn something?

5. What will you practice next?

---

12. Create Your Own Sports Problem

Choose a sport:

Write a multiplication story:

Write the multiplication sentence:

× =

Solve it:

Answer:

Explain your thinking:

13. Mini Quiz

CHAPTER 18

Recall Questions

18.1 1. What does multiplication mean?

A. Taking groups apart
B. Many equal groups combined
C. Guessing numbers
D. Racing through math

Answer:

18.2 2. What is another way to write 3 + 3 + 3 + 3?

A. 3 × 4
B. 3 + 4
C. 4 ? 3
D. 12 × 3

Answer:

CHAPTER 19

Application Questions

19.1 3. A player scores 6 points in each game for 3 games. How many points total?

Multiplication sentence:

× =

Answer:

19.2 4. A coach brings 4 bags. Each bag has 5 balls. How many balls are there?

Multiplication sentence:

× =

Answer:

CHAPTER 20

Reasoning Questions

20.1 5. Which is easier for you: repeated addition, equal groups, or skip counting? Why?

20.2 6. If you get an answer wrong, what should you do next?

14. Answer Key

CHAPTER 21

Warm-Up Practice

A. 3 × 5 = 15

B. 2 × 4 = 8

CHAPTER 22

36 Sports Multiplication Challenges

1. 5 × 4 = 20
2. 3 × 6 = 18
3. 7 × 3 = 21
4. 4 × 5 = 20
5. 8 × 5 = 40
6. 3 × 4 = 12
7. 6 × 5 = 30
8. 4 × 7 = 28
9. 2 × 6 = 12
10. 9 × 3 = 27

11. 5 × 6 = 30
12. 4 × 4 = 16
13. 6 × 4 = 24
14. 9 × 1 = 9
15. 7 × 3 = 21
16. 5 × 8 = 40
17. 4 × 5 = 20
18. 6 × 3 = 18
19. 4 × 6 = 24
20. 8 × 2 = 16
21. 5 × 6 = 30
22. 10 × 4 = 40
23. 4 × 3 = 12
24. 2 × 7 = 14
25. 6 × 5 = 30
26. 3 × 8 = 24
27. 2 × 6 = 12
28. 4 × 7 = 28
29. 9 × 3 = 27
30. 5 × 7 = 35
31. 8 × 3 = 24
32. 4 × 6 = 24
33. 3 × 5 = 15
34. 10 × 6 = 60
35. 7 × 4 = 28
36. 2 × 9 = 18

CHAPTER 23

Mini Quiz Answers

1. B. Many equal groups combined
2. A. 3 × 4
3. 6 × 3 = 18
4. 4 × 5 = 20
5. Answers will vary.
6. Answers will vary. A strong answer includes trying again, checking the groups, using a strategy, or asking for coaching.

---

15. Practice Plan for Brain Athletes

CHAPTER 24

5-Day Training Plan

24.1 Day 1: Learn the Meaning

Practice seeing multiplication as equal groups.

Goal:

"I know what multiplication means."

24.2 Day 2: Use Sports Stories

Solve 6 sports problems.

Goal:

"I can connect multiplication to real situations."

24.3 Day 3: Try Different Strategies

Use repeated addition, equal groups, and skip counting.

Goal:

"I can solve in more than one way."

24.4 Day 4: Explain Your Thinking

Tell someone how you solved a problem.

Goal:

"I can communicate my strategy."

24.5 Day 5: Reflect and Improve

Look back at mistakes and improvements.

Goal:

"I can learn from practice."

16. Memorable Takeaway

CHAPTER 25

Champions train their bodies.

CHAPTER 26

Brain Athletes train their minds.

Mistakes are not proof that you cannot learn.

Mistakes are training reps.

Understanding creates confidence.

Continue your understanding at QRU: